

An Ontology Engineering Case Study for Advanced Digital Forensic Analysis^{*}

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Abstract. Digital forensics faces some serious challenges at present. Those challenges include ever-increasing processed data volumes, heterogeneous nature of evidentiary artifacts, multiple data sources incompatible with each other, and more. Most of the commonly used forensic tools do not provide an intuitive and convenient way of accessing the data. At the same time, storage types such as relational databases cannot fully satisfy the need to store heterogeneous objects and efficiently provide access to specific properties. In this paper, we present an ontology-based approach to processing digital evidence and handling the course of digital investigation. The proposed system, named ForensicFlow, provides means of automatic artifact extraction from different origin sources, namely volatile and non-volatile memory, and reconstruction of event-artifact graphs in order to assist forensic experts in quickly and efficiently outlining the scope of an incident, and conducting an investigation.

Keywords: ontology · semantic web · digital forensics · ransomware · event reconstruction

1 Introduction

A digital forensic investigation is a highly complex and labor-intensive task requiring identification of the data sources, extraction of the data artifacts from devices having the varied OS, applications, and hardware properties, analyze and report the relevant findings [13]. A forensic process requires the utilization of different tools and techniques as there is no unique toolkit that can handle all steps. An investigator should have a substantial amount of knowledge and experience about digital forensic procedures, various technical subjects, and tools. Despite various shortcomings, the existing forensic tools usually provide a working solution for collecting and presenting the data artifacts individually to the investigator. However, the analysis and reporting phases require deducing sophisticated correlations between these artifacts, which is mostly not fulfilled

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in the current forensic technology. Lack of this data analytic perspective in the forensic tools causes a huge cognitive burden on investigators, prolongs the investigation duration, and creates a huge dependence on the technical capabilities of investigators in an area that mostly lacks the necessary talents. All these implications reflect the need for effective knowledge management in digital forensics [5].

A body of digital forensic research under the topic of event reconstruction has addressed this analytical gap by using timeline analysis. This analysis mainly filters the system and network events based on the timing criteria and creates chronological order for the investigator [7]. Although time is a very significant parameter in revealing relations, it does not grasp all aspects implicit in the sequences of artifacts that reflect the user or system behavior in question or is not sufficient to discriminate the relevant artifacts from the enormous size of artifact pool that can be observed in any evidence. Complex cases such as malware investigations necessitate linking the user activity with some system activities by using semantic relations in addition to the time variable. Another limitation of the studies utilizing timeline analysis is that they only concentrate on the artifacts obtained from non-volatile memory sources.

Ontologies allow the logical representation of the knowledge of a certain domain by utilizing semantic relationships between class individuals. They act as a common language between experts but can also instrument transferring the created knowledge to machine-readable formats and establish a baseline for reasoning. Therefore, digital forensics can significantly benefit from ontologies and semantic frameworks as this domain mainly revolve around the application of expert knowledge. Although a body of research has explored the utilization of ontologies in the digital forensics domain [14], still, an important applied research gap exists that can be addressed by forming a more advanced reasoning system for better investigation capability in complex cases or security incidents.

In this study, we created a semantic web framework consisting of an ontology, graph model, and relevant semantic queries for the analysis of digital forensic cases. In order to populate the ontology model with forensic data obtained from volatile and non-volatile memory sources, we complemented the framework with data extraction tools. We run the forensic investigation cycle for a simulated ransomware case to demonstrate the feasibility of our solution. The main contribution of our study is the development of a comprehensive reasoning framework that utilizes a variety of resources for resolving advanced forensics cases.

2 Background Information about Digital Forensic Processes

Major principles of digital forensics evidence handling aim for adequate preservation and analysis of evidence so that it is admissible in the court. Several process models have been developed, but the majority of them share the same steps:

1. Identification and Seizure. This is a process of identifying potential sources of evidentiary data that relate to a crime scene and seizing these sources.
2. Acquisition or Preservation. This stage involves copying evidence sources in such a way that they may be later accounted for in court.
3. Analysis. After evidence sources have been acquired they are carefully analyzed by a forensic expert. The primary objective of the analysis is to obtain evidence to support or disprove a hypothesis carried out by an investigation. During this process, an investigator usually applies a wide set of different methods, tools, and techniques due to the heterogeneous nature of artifacts analyzed.
4. Presentation and Reporting. After conclusions are made a formal expert report is compiled to be later used in the court or other institutional settings.

Digital forensic processes usually have the post-mortem character, meaning that they target cases that have already fully or partially completed.

3 Literature Review

Timeline analysis studies could be classified as super, micro, or nano based on the time and data source coverage [7]. Super timeline covers the widest period enabling the investigator to form various queries [8] whereas micro or nano types address more limited timeline and source variety [3]. log2timeline is a widely-known forensic tool [9] that extracts events from different sources (e.g., images, files), creates a plaso storage file that includes the whole timeline and event content, and enables to query via different command-line tools such as pininfo and psort [1].

The high-level activity patterns (e.g., USB stick insertion) have been derived from system-level events to automatize some forensic efforts by using pre-determined rules [10]. Another study enables to do a correlation of various events retrieved from various resources including disk and memory images [4]. However, these researches do not utilize a common and understandable framework for creating the rules, limiting the extensibility and applicability of the contribution.

A review of the ontology engineering studies in the digital forensic domain is given in [14]. Various taxonomies have been developed to conceptualize the domain and classify the relevant terms in this domain. For instance, a very general taxonomy regarding the technological and professional aspects is provided in [2] whereas a more specific one categorizes the technical forensic subjects and related evidence sources [12].

A general ontology for digital investigation is proposed in [11]. This study details only the analysis of the Windows registry. In a more comprehensive study, an ontology is populated by log2timeline outputs and used for deriving inferences regarding the missing user data [6]. All these contributions usually focus on artifacts about user events extracted from non-volatile memory resources, which makes them not suitable for investigating the more complex cases requiring the comprehension of relations between the user and system-level events coming from all resources including volatile-memory.

4 Methodology

Our framework developed in this study addresses the analysis step of the digital forensic process. We formed an ontology for the forensic artifacts that can be obtained from volatile and non-volatile memory resources. This ontology constitutes the core conceptualization component of the system proposed. We utilized SPARQL to query semantic relationships. The system consists of the following major layers: data sources, extraction modules, the analyzer, knowledge base, and knowledge access interface (see Fig. 1). Extraction modules can have any implementation but the output must comply with the following rules: it should produce instances of ontology-defined classes and a mapping profile for correlations with other class instances that may or may not be found in other sources. The output of extractors is then fed to the analyzing module that performs instances correlation using object properties available from the schema. The result of the analyzer is instantiated ontology in the form of an RDF triplestore. A SPARQL query interface then allows retrieval of the stored knowledge.

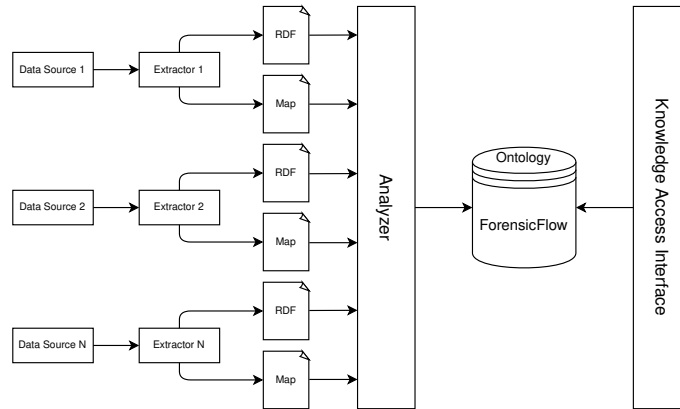


Fig. 1. Overview of the system.

ForensicFlow Ontology Design W3C Web Ontology Language (OWL) was used for the ontology design as one of the most widely used and supported by a powerful technology stack. The ease of ontologies merging, allowing a combination of different knowledge stores, was an additional factor to decide in favor of OWL. To represent the knowledge and achieve previously discussed goals the following criteria needed to be met:

- Uniform individuals representation to allow new data sources and artifact types to be introduced, but share the same interface.
- Storing of artifact-specific data. All the type-specific knowledge needs to be carefully preserved inside the individuals.

- The ontology needs to have a pluggable architecture to allow easy adoption of any new data sources or artifact types that may appear in the future.

To achieve this goal two base classes were defined: *Event* and *ForensicArtifact*. These two major classes define a common interface by the principles of polymorphism and object-oriented design so that any sub-classes derived from them could be queried in the same manner.

The *Event* base class represents a generic event in a crime scene, e.g. an executable was run or a web page was visited. It has only one data property of its own which is event time and one generic object property that is *hasArtifact*. Those properties get inherited by all of its sub-classes. The many children of the event class may include any events that happen in the system and any data accompanying them. The exact system events are then derived from this class introducing knowledge specific to them. Object properties for Event sub-classes are generated by extending the *hasArtifact* property of the parent class. A good example of such a link can be a file download event linking to the file's metadata extracted from the MFT record.

Generic *ForensicArtifact* class includes information about the data source it was extracted from, the name of the investigator, and any particulars regarding the extraction procedure that may be relevant to those trying to replicate the process. There are several subgroups of artifacts defined to structure them better. The groups logically divide artifacts by nature, e.g. network, file system, processes, etc. Each group then defines the exact artifact type or introduces further grouping.

The described organization creates a versatile system of connection between events and artifacts involved in them with a universal interface for querying objects and related connections.

Artifacts Extraction and Ontology Instantiation To populate the ontology with knowledge about crime scene artifacts and their mutual interconnections the expert is expected to run a set of automatic extraction tools against the evidence sources in possession. Currently, two data sources are supported: hard drive images and volatile memory snapshots. The advantage of mixed-origin data storage is that the investigator can quickly and conveniently get a different perspective of the same data or events or augment one source with evidence from another. Each tool extracts relevant pieces of information and serializes them in OWL format as ontology individuals. At the same time, it creates a mapping profile for the analyzer to digest later if artifact linking is suspected. The mapping profile is a set of links to potential individuals that could be discovered by other extractors and is used by the analyzer to create semantic relations. To address the excessive data issue mentioned earlier all extractors operate only on a subset of data that is relevant to the investigation. Currently, there are three tools developed in order to test out our hypothesis:

- Volatility plug-in operates on a volatile memory (RAM). The extracted information includes running processes information, executable path, open network connections, etc.

- Log2Timeline handler extracts valuable information from L2T reports such as prefetch events, browser activity, registry edits, etc. Data generated by this software populates the Events domain of the ontology.
- Autopsy plug-in extracts file-related data from the disk image, taking into account mapping profiles generated in previous steps to comprise a full map of forensic artifacts. At the moment the data extracted by the plug-in includes basic statistics such as file size, mime type, and location, as well as some advanced ones like entropy anomalies.

The resulting OWL individuals and mappings are then fed to the analyzer that populates the ontology with instances and their relations according to the schema. Due to the rules defined in each extractor, only relevant data is fetched from original sources that dramatically reduces the size of analyzed information.

5 Experimentation and Results

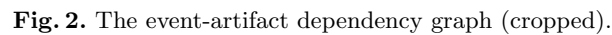
This section covers an experimental case study and covers some preliminary conclusions. The primary target here is to demonstrate the advantages of the proposed method over traditionally accepted forensic practices and outline the capabilities of the system in a life-like scenario. The simulated attack represents a system where a malicious application (ransomware) is first downloaded from a network resource and later executed following the encryption of some of the user's files. The aim of the investigator is to quickly and conveniently identify the cause, the source, and the damage caused by an attack.

A Windows 7 virtual machine acts as a target system. Some typical user files are populated in usual locations inside user space. A malicious application is uploaded to a website in advance. No antivirus software, aside from built-in Windows Defender is present in the system. The test ransomware used in the experiment when executed goes over the user directory and selectively encrypts files with AES cipher by a predefined extensions mask.

The attack begins with the user visiting a "malicious" web page and downloading the ransomware executable through the Firefox web browser to a standard download location. The malicious application is then executed and the user files get encrypted after some time. When the attack is done a ransom message is saved to the user desktop as a text file. The application stays resident and tracks any further file changes encrypting newly created files.

At this stage, the investigator moves in and performs a memory dump in order to save any volatile information that is still present in the system. The system is then shut down and a forensic disk image is taken using FTK Imager. The investigator then runs our modules against these sources in order to instantiate an ontology that is ready for analysis.

The standard forensic procedure, in this case, would include WinPrefetch events lookup. These events indicate an application execution occurrence. When suspicious execution has been identified events surrounding that execution are being manually analyzed in an attempt to figure out the context. Any traces



By utilizing the proposed ontology a single SPARQL query for WinPrefetch events reveals 19 instances from which only two seem suspicious since they were performed from the userspace. From the Protege graph view, these events are searched revealing the full graph of dependent artifacts (see Fig. 2). The graph reveals some crucial facts about the attack. From the suspicious WinPrefetch event (1) we discover the executable file in question (2) and all of its related events and linked artifacts. The file first appears in the Firefox web browser download event (3) that reveals the network resource of origin (4). The file is then recorded by the file system as written to disk in the user download folder(5). The next event it appears in is the execution event we started from (1). Lastly, the file is traced back to the exact process in memory that is still running (6), making it possible to extract the memory context and possibly the encryption key.

To perform a similar analysis using only standard tools the examiner would have to switch between at least 3-4 different tools performing manual searches for each artifact, noting down the correlations between them, and trying to reconstruct the flow of events. We would also like to note that currently there are no tools on the market that correlate hard disk data and memory extracted information. Thus the target goals have been met to a significant extent. An ontology for a crime scene was developed. It allows for convenient storing and querying of event-artifact-event instances making it much easier for an investigator to outline the scope of events surrounding certain actions. An extensible interface for the knowledge store population has been introduced allowing for any third

party to enrich the current tool-set. The extraction modules architecture implies significant initial data reduction making it easier to analyze.

6 Conclusion and Future Work

In this paper, we proposed an ontology-based approach to accurately and effectively store digital crime scene evidence artifacts. Our ontology is accompanied by a set of evidence extraction modules to cover different evidence sources, an analyzer engine that is responsible for the mutual correlation of artifacts, and a set of SPARQL queries to fetch the data out of the ontology. The extraction modules take care of excessive data reduction by applying predefined rules that fetch only the data relevant to crime scene reconstruction. We consider several topics as parts of the future work. First of all, we want to extend the scope of data sources to include not only multiple origins for one device but systems of multiple devices connected in one way or another. This approach would allow automating investigations in a distributed environment such as the Internet of Things. Another point we would like to address is the validation with actual practitioners including law enforcement and private experts.

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